BUSINESS RESEARCH METHODS

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RESEARCH

- **Re ---------- Search**

- **Re** means once more, again, anew
- **Search** means try to find something by looking or otherwise seeking carefully and thoroughly.

- **Research** is an **Organized** and **Systematic** way of **Finding Answers** to **Questions**.

RESEARCH

- **Systematic:** because there is a definite set of procedures and steps which you will follow. There are certain things in the research process which are always done in order to get the most accurate results.
- **Organized:** in that there is a structure or method in going about doing research. It is a planned procedure, not a spontaneous one. It is focused and limited to a specific scope.
- **Finding Answers** : is the end of all research. Whether it is the answer to a hypothesis or even a simple question, research is successful when we find answers. Sometimes the answer is no, but it is still an answer.
- **Questions** :are central to research. If there is no question, then the answer is of no use. Research is focused on relevant, useful, and important questions. Without a question, research has no focus, drive, or purpose.

BUSINESS RESEARCH

Research in these fields is also broadly referred to as life and physical science research and social science research. Business education covers both Commerce and Management, which are part of Social sciences.

- **Business research is an organised, systematic and objective process of gathering ,recording, and analysing and interpreting the data that provide information to make accurate business decisions.**
- **Business research may be defined as a systematic inquiry whose objective is to provide the information that will allow managerial problems to be solved.**

BUSINESS RESEARCH

Management problem : any problem or opportunity that requires a **management decision**.

- Problem

  A condition that occurs when some aspect of organizational performance is less than desirable.
- Opportunity

  Any situation that has the potential to provide additional beneficial outcomes.
- Decision making

  The process through which managers and leaders identify and resolve problems and capitalize on opportunities.

BUSINESS RESEARCH

Who does business research ?

- Academics
- Students
- Government and Commercial organisation
- Consultants
- Managers

BUSINESS RESEARCH

Business research is a broad term which covers many areas.

- Accounting
- Finance
- Marketing
- Human Resources
- Operation Research
- Technology
- Agribusiness Management
- Healthcare Management

BUSINESS RESEARCH

- The key objective of any business research is to provide accurate and timely information to the top management, so that they can make effective decisions.
- Research studies evolve through a series of steps, each representing the answer to key questions.

➢ Why should we do research ?

➢ What research should be done ?

➢ How should the research be designed to achieve the research objectives ?

➢ How will it be analysed and interpreted the data?

➢ What will we do with the research ?

TYPES OF RESEARCH

- **1. Basic Research:** It is also known as pure or fundamental research.

  - This research is mainly conducted to increase knowledge base. It is driven purely by interest and a desire to expand our knowledge.
  - This type of research tends not to be directly applicable to the real world in a direct way but enhances our understanding of the world around us.
  - Basic research generates new ideas, principles and theories indifferent fields.
  - It is sometimes implicitly said that basic research doesn’t have practical applications.

2. Applied Research:

- Applied research is mainly related with solving practical problems rather than focusing on knowledge expansion.
- It is mainly used to find solutions to problems which occur daily and develop new innovative technologies.
- The main aim of applied research is to provide better technologies for humans to enhance their standard of living.

3. Quantitative Research:

- Quantitative research is generally related with the positivist concept.
  - It usually involves collecting and converting data into numerical form so that statistical calculations can be made, and conclusions drawn.
  - Objectivity is very vital in quantitative research.
  - Therefore, researchers try to avoid their own presence, behavior or attitude affecting the results (e.g., by changing the circumstances being studied or causing participants to behave differently).
  - They also examine their methods and results for any possible bias.
  - The aim of quantitative research is to develop mathematical models, theories related to phenomenon.
  - Quantitative research is mainly used in social sciences.

4.Qualitative Research:

- Qualitative research is the approach usually related with the social constructivist concept which emphasizes the socially constructed nature of reality.
- It is about recording, analyzing and attempting to reveal the in depth meaning and significance of human behavior and experience, including conflicting beliefs, behaviors and emotions.
- The qualitative method tries to answer why and how of decision-making rather than what and when.
- The approach to data collection and analysis is logical but allows for greater flexibility than in quantitative research.
- Data is collected in textual form based on observation and communication with the participants

5.Descriptive Research:

- Descriptive research is used to describe characteristics of an observable fact being studied.
- Descriptive studies are structured in such a way that they cannot be changed frequently, so it can be said that they are rigid in nature.
- They cannot identify cause and effect relationship between variables.
- Descriptive research answers questions such as who, when, where, what and how.
- This type of research describes what exists and may help to reveal new facts and meanings.
- The purpose of descriptive research is to **observe, describe and document**.

6.Exploratory Research:

- Exploratory research is carried out for a problem that has not been clearly defined.
- The main aim of this research is to gather initial information which helps to define problems and recommend hypothesis.
- Exploratory research helps to settle on the best research design, data collection method and selection of subjects.
- Exploratory research often relies on secondary research such as reviewing available literature, or qualitative approaches such as informal discussions with consumers, employees, management or competitors, and more formal approaches through in- depth interviews, focus groups, projective methods, case studies or pilot studies.
- Exploratory research can mainly be conducted when researchers lack clear idea of the problem.
- The results of exploratory research are not generally useful for decision-making, but they can provide major insight into a given situation.

7. Historical Research:

- It is defined as the type of research that examines past events or combinations of events to arrive at an account of what has happened in the past.
- Historical research is carried out to discover the unknown; answer questions, recognize the relationship that the past has to the present; record and assess activities of individuals, agencies, or institutions; and assist in understanding the culture in which we live.
- Historical research can exhibit patterns that occurred in the past and over time which can facilitate us to see where we came from and what kinds of solutions we have used in the past.
- We usually will notice that what we do today is expressly rooted in the past. Historical research involves the process of collecting and reading the research material collected and writing the document from the data collected.

8. Experimental Research:

- It is commonly used in sciences such as sociology and psychology, physics, chemistry, biology, medicine, etc.
- It is a collection of research designs which use manipulation and controlled testing to understand fundamental processes.
- Usually, one or more variables are manipulated to establish their effect on a dependent variable.
- Experimental Research is mainly used when: there is time priority in a causal relationship (cause precedes effect) or there is uniformity in a causal relationship (a cause will always lead to the same effect) or the magnitude of the correlation is great.
- Experimental research is important to society as it helps us to improve our daily lives.
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Note: Researchers use various research methods to collect relevant data so that business enterprises can make wiser decisions. There are two main types of methods to carry out business research.

9. Causal research

- Causal research, also known as explanatory research is conducted to identify the extent and nature of cause-and-effect relationships.
- Causal research can be conducted to assess impacts of specific changes on existing norms, various processes etc. Causal studies focus on an analysis of a situation or a specific problem to explain the patterns of relationships between variables.

Basis of difference	Causal research	Exploratory research
Amount of uncertainty characterizing decision situation	Clearly defined	Highly ambiguous
Key research statement	Research hypotheses	Research question
When conducted?	Later stages of decision making	Early stage of decision making
Usual research approach	Highly structured	Unstructured
	‘Will consumers buy more products in a blue package?’ Which of two advertising campaigns will be more effective?’	‘Our sales are declining for no apparent reason’ ‘What kinds of new products are fast-food consumers interested in?’

10.Theoretical research

- Theoretical research enables you to explore and discuss a research object based on your use of abstract theoretical structures and philosophical concepts.
- You cannot, in theoretical research, directly observe the research object. The aim of your research is to define and outline conceptual models, explanations and structures of the chosen topic with the help of research literature.

11.Empirical research

Empirical research is defined as any research where conclusions of the study is strictly drawn from concretely empirical evidence, and therefore “verifiable” evidence.
This empirical evidence can be gathered using quantitative market research and qualitative market research methods

12. Cross-sectional research

- It involves using different groups of people who differ in the variable of interest but share other characteristics, such as socioeconomic status, educational background, and ethnicity.
- Cross-sectional research studies are often used by researchers studying developmental psychology.
- The participants in this type of study are selected based on variables of interest. Cross-sectional studies are often used in developmental psychology, but this method is also utilized in many other areas including social science and education.

13.Time series research

- Time series analysis is a statistical technique that deals with time series data, or trend analysis.
- Time series data means that data is in a series of time periods or intervals.
- Time series data: A set of observations on the values that a variable takes at different times.

	Qualitative research	Quantitative research
Focus	Exploring ideas or formulating hypotheses/theories	Testing hypotheses or theories
Analysis	Summarizing, categorizing, interpreting	Math and statistical analysis
Expressed in	Words	Numbers, graphs, tables, fewer words
Sample	Few respondents	Many respondents
Questions	Open-ended	Close-ended or multiple choice
Characterized by	Understanding, context, complexity, subjectivity	Testing, measurement, objectivity, replicability

BUSINESS RESEARCH PROCESS



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BUSINESS RESEARCH PROCESS

- The research begins with a problem and proceeds through data collection and analysis to the written report.
- The research process provides a systematic, planned approach to the research project.

i) Selection of a research problem or researchable area.

ii) Literature review

iii)Definition and specification of the research problem more clearly.

iv)Formulation of research hypothesis or at least research objectives.

v)Preparing the research design, i.e., identification of the sources of data and sample design.

BUSINESS RESEARCH PROCESS

vi)Creation and construction of data collection instruments like Questionnaire, Scales etc.

vii)Pilot testing of the instruments and their possible revision.

viii)Formal acquisition of data and information, through survey, observation, interview etc.

ix) Processing and analysis of the data.

x)Interpretation of the data and formal write up i.e., reporting.

xi) Management decision

Research Design

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Research Design: Definition

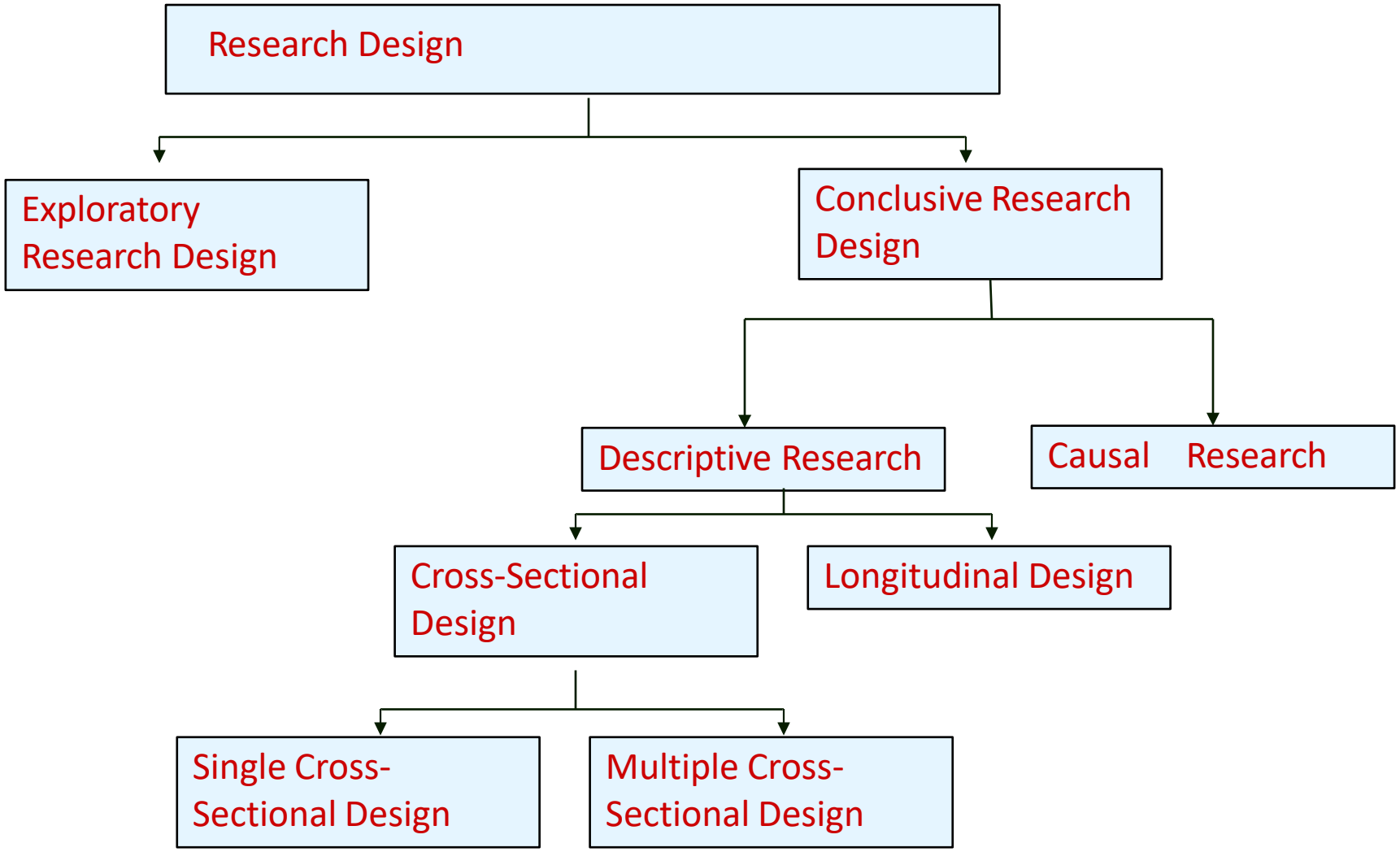
- A **research design** is a framework or blueprint for conducting the marketing research project. It details the procedures necessary for obtaining the information needed to structure or solve marketing research problems.

Components of a Research Design

- Define the information needed
- Design the exploratory, descriptive, and/or causal phases of the research
- Specify the measurement and scaling procedures
- Construct and pretest a questionnaire (interviewing form) or an appropriate form for data collection
- Specify the sampling process and sample size
- Develop a plan of data analysis

1) Statement of Problem
2) Need for the Study
3) Review of Previous Studies
4) Objectives of Study
5) Formulation of Hypothesis
6) Operational Definitions
7) Scope of the Study
8) Sources of Data
9) Sampling Design
10) Tools & Techniques

A Classification of Marketing Research Designs



Exploratory & Conclusive Research Differences

Table 3.1

	Exploratory	Conclusive
Objective:	To provide insights and understanding.	To test specific hypotheses and examine relationships.
Characteristics:	Information needed is defined only loosely. Research process is flexible and unstructured. Sample is small and non-representative. Analysis of primary data is qualitative.	Information needed is clearly defined. Research process is formal and structured. Sample is large and representative. Data analysis is quantitative.
Findings/ Results:	Tentative.	Conclusive.
Outcome:	Generally followed by further exploratory or conclusive research.	Findings used as input into decision making.

A Comparison of Basic Research Designs

	Exploratory	Descriptive	Causal
Objective:	Discovery of ideas and insights	Describe market characteristics or functions	Determine cause and effect relationships
Characteristics:	Flexible, versatile	Marked by the prior formulation of specific hypotheses	Manipulation of one or more independent variables
	Often the front end of total research design	Preplanned and structured design	Control of other mediating variables
Methods:	Expert surveys Pilot surveys Secondary data: qualitative analysis Qualitative research	Secondary data: quantitative analysis Surveys Panels Observation and other data	Experiments

Uses of Exploratory Research

- Formulate a problem or define a problem more precisely
- Identify alternative courses of action
- Develop hypotheses
- Isolate key variables and relationships for further examination
- Gain insights for developing an approach to the problem
- Establish priorities for further research

Methods of Exploratory Research

- Survey of experts
- Pilot surveys
- Secondary data analyzed in a qualitative way
- Qualitative research

Use of Descriptive Research

- To describe the characteristics of relevant groups, such as consumers, salespeople, organizations, or market areas.
- To estimate the percentage of units in a specified population exhibiting a certain behavior.
- To determine the perceptions of product characteristics.
- To determine the degree to which marketing variables are associated.
- To make specific predictions.

Methods of Descriptive Research

- Secondary data analyzed in a quantitative as opposed to a qualitative manner
- Surveys
- Panels
- Observational and other data

Data Collection
and
Questionnaire Design

DATA

Information about a particular phenomenon.

DATA TYPES

Primary Data

Secondary Data

Primary Data

- Primary data means original data that have been collected specially for the purpose in mind.
- It means when an authorized organization or an investigator or an enumerator collects the data for the first time himself or with the help of an institution or an expert then the data thus collected are called primary data.
- Primary data is important for all areas of research because it is unvarnished information about the results of an experiment of observation.
- It is like the eyewitness testimony at a trial.
- No one has tarnished it or spun it by adding their own opinion or bias so it can form the basis of objective conclusions.

Advantages of Primary Data

- Degree of accuracy is quite high.
- It does not require extra caution
- It depicts the data in great details
- Primary source of data collection frequently includes definitions of various terms and units used.
- For some investigations, secondary data are not available .

Disadvantages of Primary Data

- Collection of data requires a lot of time
- It requires lot of finance
- In some enquires it, is not possible to collect primary data
- It requires a lot of labour.

Secondary Data

- Secondary data are data that have been collected for another purpose and where we will use statistical method with the Primary Data.
- It means that after performing statistical operations on Primary data the results become known as Secondary Data.
- Secondary data is the data that have been already collected by and readily available from other sources.
- Such data are cheaper and more quickly obtainable than the primary data and also may be available when primary data cannot be obtained at all.

Advantages of Secondary Data

- It is economical &saves efforts and expenses
- It helps to make primary data collection more specific since with the help of secondary data.
- We are able to make out what are the gaps and deficiencies and what additional information needs to be collected
- It helps to improve the understanding of the problem.
- It provides a basis for comparison for the data that is collected by the researcher.

Disadvantages

- Secondary data is something that seldom fits in the framework of the marketing research factors. Reasons for its non-fitting are:
  - Unit of secondary data collection – suppose you want information on disposable income, but the data is available on gross income. The information may not be same as we require
  - Class boundaries may be different when units are same.
  - Thus, the data collected earlier is of no use to you
- Accuracy of secondary data is not known.
- Data my be outdated.

Difference between Primary & Secondary Data

Primary Data	Secondary Data
• Needs more funds • Investing Agency collects the data • Requires longer time for collection • More reliable and suitable to the enquiry because the investor himself collects it • Requires elaborate organization • No extra precautions are required	• Needs comparatively less funds • Some other investing agency collects it for its own use • Requires less time for collection • Less reliable and suitable as someone else does collection which may not set the purpose • No need of any organization set-up

Primary Data Collection Method

Primary data can be collected through

1. Observation Method
2. Survey Method
3. Questionnaire
4. Schedule

Observation Method

- The observation method involves human or mechanical observation of what people actually do or what events take place during a buying or consumption situation. “Information is collected by observing process at work”. The following are a few situations:
  - Service stations – pose as a customer, go to a service station and observe.
  - To evaluate the effectiveness of display of Dunlop Pillow Cushions – in a departmental store, observer notes: a) How many pass by; b) how many stopped to look at the display and c) How many decide to buy.
  - Super Market – which is the best location in the shelf? Hidden cameras are used.
  - To determine typical sales arrangement and find out sales enthusiasm shown by various salesmen – normally, this is done by an investigator using a concealed tape-recorder.

Types of Observation Method

- Participant
- Non-participant

Advantages of Observation Method

- If the researcher observes and records events, it is not necessary to rely on the willingness and ability of respondents to report accurately
- The biasing effect of interviews is either eliminated or reduced. Data collected by observation are, thus, more objective and generally more accurate

Disadvantages of Observation Method

- The most limiting factor in the use of observation method is the inability to observe things such as attitudes, motivations, customers/consumers state of mind, their buying motives and their images.
- It also takes time for the investigator to wait for a particular action to take place
- Personal and intimate activities, such as watching television late at night, are more easily discussed with questionnaires than they are observed.
- Cost is the final disadvantage of observation method. Under most circumstances, observational data are more expensive to obtain than other survey data.
- The observer has to wait doing nothing, between events to be observed. The unproductive time is an increased cost.

Survey Method

- The survey method is the technique of gathering data by asking questions to people who are thought to have desired information.
- A formal list of questionnaire is prepared.
- Generally, a non-disguised approach is used.
- The respondents are asked questions on their demographic interest opinion.

Advantages of Survey Method

- As compared to other methods (direct observation, experimentation) survey yield a broader range of information.
- Surveys are effective to produce information on socio-economic characteristics, attitudes, opinions, motives etc. and to gather information for planning product features, advertising media, sales promotion, channels of distribution and other marketing variables.
- Questioning is usually faster and cheaper than observation
- Questions are simple to administer
- Data is reliable
- The variability of results is reduced
- It is relatively simple to analyze, quote and interrelate the data obtained by survey method.

Disadvantages of Survey Method

- Unwillingness of respondents to provide information – this requires salesmanship on the part of the interviewer.
- The interviewer may assure that the information will be kept secret or apply the technique of offering some presents.
- Inability of the respondents to provide information – this may be due to:
  - Lack of knowledge
  - Lapse of memory
  - Inability to identify their motives and provide 'reason why?' for their actions

•Human biases of the respondents are there, for e.g. 'Ego'

•Semantic difficulties are there – it is difficult, if not impossible, to state a given question in such a way that it will mean exactly same thing to each respondent.

•Similarly two different wording of the same question will frequently produce quite different results.

Techniques of Survey Methods

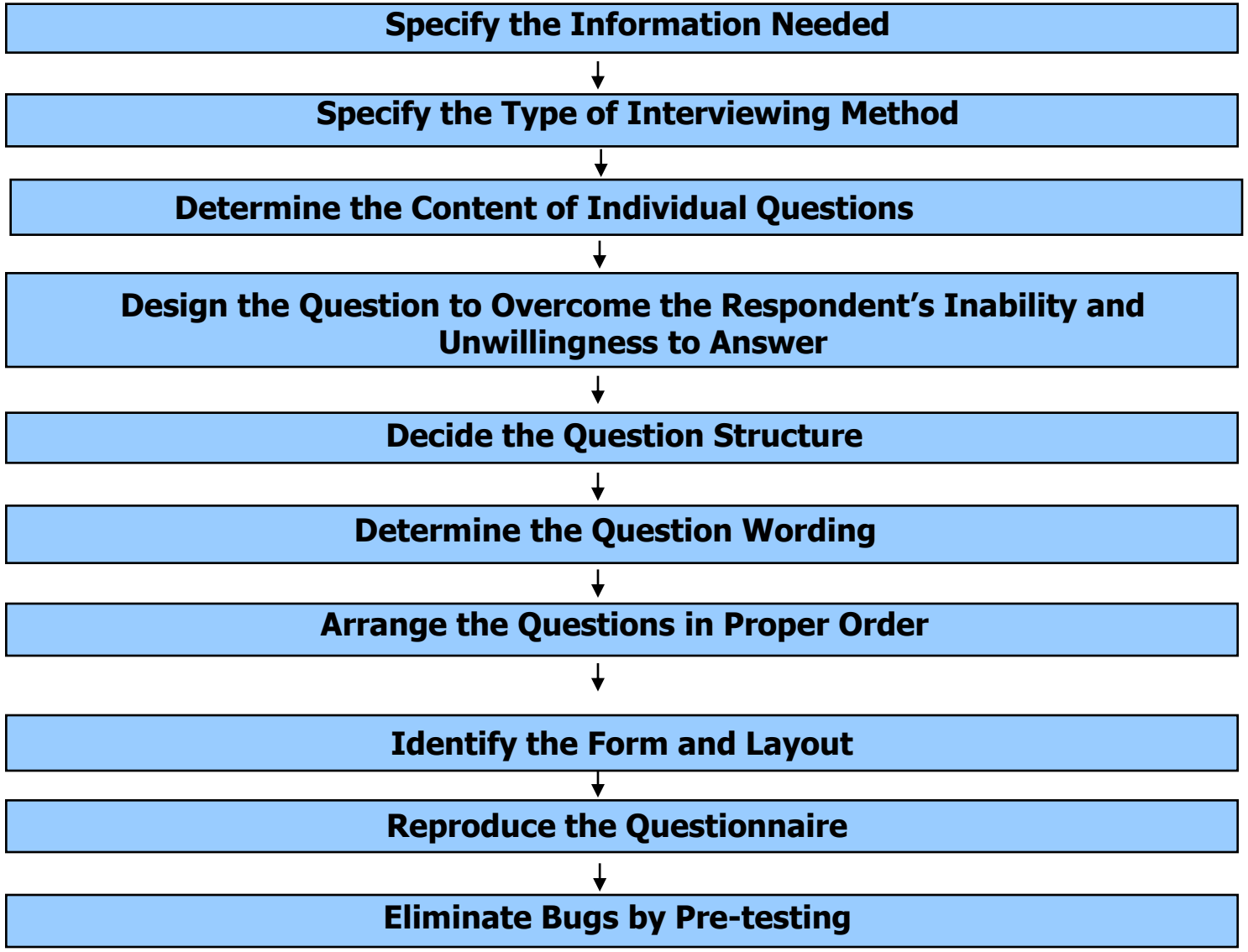
There are mainly four methods by which we can collect data through the survey method.

- Telephonic Interview
- Personal Interview
- Mail Interview
- Electronic Interview

Collection of Data through Questionnaires

- A questionnaire is a formalized set of questions for obtaining information from respondents.
- It must translate the information needed into a set of specific questions that the respondents can and will answer.
- A questionnaire must uplift, motivate, and encourage the respondent to become involved in the interview, to cooperate, and to complete the interview
- A good questionnaire should not be tool lengthy.
- Simple English should be used and the question shouldn’t be difficult to answer.
- A good questionnaire requires sensible language, editing, assessment, and redrafting

Questionnaire Design Process



Survey Design(Questionnaire Design Process)

1. State the information required: This will depend upon the nature of the problem, the purpose of the study and hypothesis framed. The target audience must be concentrated on.

2. State the kind of interviewing technique: Interviewing method can be telephone, mails, personal interview or electronic interview.

3. Decide the matter/content of individual questions: There are two deciding factors for this:

a) Is the question significant? – Observe contribution of each question. Does the question contribute for the objective of the study?

b) Is there a need for several questions or a single question? – Several questions are asked in the following cases:

- When there is a need for cross-checking.
- When the answers are ambiguous
- When people are hesitant to give correct information.

4. Overcome the respondents' inability and unwillingness to answer:

- The Respondents may be unable to answer the questions because of the following
  - The respondent may not be fully informed
  - The respondent may not be remember
  - He may be unable to express or articulate
  - The Respondent may be unwilling to answer due to:
  - There may be sensitive information which may cause embarrassment or harm the respondent's image.
  - The respondent may not be familiar with the genuine purpose
  - The question may appear to be irrelevant to the respondent.
  - To overcome the respondents unwillingness to answer:
  - Place the sensitive topics at the end of the questionnaire
  - Preface the questions with a statement
  - Use the third person technique (For ex: - Mark needed a job badly and he used wrong means to get it – is it right? Different people will have different opinions depending upon the situation)

5.Decide on the structure of the Question

Questions can be of two types:

a) Structured questions: These specify the set of response alternative and the response format. These can be classified into multiple choice questions, dichotomous questions and scales

b) Unstructured questions: These are also known as open-ended question. No alternative are suggested and the respondents are free to answer these questions in any way they like.

6.Determine the question language/phrasing:

1.Define the issue in terms of who, what, when, where, why, and way (the six Ws).

2.Use ordinary words. Words should match the vocabulary level of the respondents.

3.Avoid ambiguous words: usually, normally, frequently, often, regularly, occasionally, sometimes, etc.

4.Avoid leading questions that clue the respondent to what the answer should be.

5.Respondent should not have to make generalizations or compute estimates.

6.Use positive and negative statements.

7.Properly arrange the questions:

1. The opening questions should be interesting, simple, and non-threatening.
2. Qualifying questions should serve as the opening questions.
3. Basic information should be obtained first, followed by classification, and, finally, identification information.
4. Difficult, sensitive, or complex questions should be placed late in the sequence.
5. General questions should precede the specific questions.
6. Questions should be asked in a logical order.
7. Branching questions should be designed carefully to cover all possible contingencies.

8.Recognize the form and layout of the questionnaire:

- Divide a questionnaire into several parts.
- The questions in each part should be numbered, particularly when branching questions are used.
- The questionnaires should preferably be pre-coded.
- The questionnaires themselves should be numbered serially.

9.Reproduce the questionnaire:

- The questionnaire should be reproduced on good-quality paper and have a professional appearance.
- Questionnaires should take the form of a booklet rather than a number of sheets of paper clipped or stapled together.
- Each question should be reproduced on a single page (or double-page spread).
- Vertical response columns should be used for individual questions.
- Grids are useful when there are a number of related questions which use the same set of response categories.
- The tendency to crowd questions together to make the questionnaire look shorter should be avoided.
- Directions or instructions for individual questions should be placed as close to the questions as possible.

10.Pre-test the questionnaire:

- Pretesting should be done always.
- All aspects of the questionnaire should be tested, including question content, wording, sequence, form and layout, question difficulty, and instructions.
- The respondents in the pretest should be similar to those who will be included in the actual survey.
- Begin the pretest by using personal interviews.
- Pretest should also be conducted by mail or telephone if those methods are to be used in the actual survey.
- A variety of interviewers should be used for pretests.
- The pretest sample size is small, varying from 15 to 30 respondents for the initial testing.
- Use protocol analysis and debriefing to identify problems.
- After each significant revision of the questionnaire, another pretest should be conducted, using a different sample of respondents.
- The responses obtained from the pretest should be coded and analyzed.

Difference b/w Questionnaire / Schedule

- In practical life, schedule or questionnaires have same meaning.
- But in theoretical aspect, difference is made between a schedule or questionnaire.
- If an investigator himself fills the form by asking face to face questions to informants, it will be called a schedule, but on the other hand, if the same work is done by sending the form to the information and information himself fills the form and sends it back, the form is called questionnaire.

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The important points of differences are as under:

1. Questionnaire is filled by informants but schedule is filled by enumerators, who are specially appointed for this purpose.
2. In questionnaire, if the questions are complex, there are chances of non-response but in schedule there is no chance of non-response
3. Questionnaire is the list or series of questions prepared to be asked but schedules is scheduled events or plans to be executed on specific days or dates
4. To collect data through questionnaire is relatively cheap and economical but through schedules is relatively more expensive.
5. In case of questionnaire, it is not always clear as to who replies, but in case of schedule the identity of respondent is known.

Measurement and Scaling

Measurement

Measurement means assigning numbers or other symbols to characteristics of objects according to certain pre-specified rules.

One-to-one correspondence between the numbers and the characteristics being measured.

The rules for assigning numbers should be standardized and applied uniformly.

Rules must not change over objects or time.

Scaling

Scaling involves creating a continuum upon which measured objects are located.

Consider an attitude scale from 1 to 100. Each respondent is assigned a number from 1 to 100, with 1 = Extremely Unfavorable, and 100 = Extremely Favorable. Measurement is the actual assignment of a number from 1 to 100 to each respondent. Scaling is the process of placing the respondents on a continuum with respect to their attitude toward department stores.

Primary Scales of Measurement

Scale		
Nominal	Numbers Assigned to Runners	
Ordinal	Rank Order of Winners	 **Third place** **Second place** **First place**
Interval	Performance Rating on a 0 to 10 Scale	**8.2** **9.1** **9.6**
Ratio	Time to Finish, in Seconds	**15.2** **14.1** **13.4**

Nominal Scale

- The numbers serve only as labels or tags for identifying and classifying objects.
- When used for identification, there is a strict one-to-one correspondence between the numbers and the objects.
- The numbers do not reflect the amount of the characteristic possessed by the objects.
- The only permissible operation on the numbers in a nominal scale is counting.
- Only a limited number of statistics, all of which are based on frequency counts, are permissible, e.g., percentages, and mode.

Ordinal Scale

A ranking scale in which numbers are assigned to objects to indicate the relative extent to which the objects possess some characteristic.

Can determine whether an object has more or less of a characteristic than some other object, but not how much more or less.

Any series of numbers can be assigned that preserves the ordered relationships between the objects.

In addition to the counting operation allowable for nominal scale data, ordinal scales permit the use of statistics based on centiles, e.g., percentile, quartile, median.

Interval Scale

- Numerically equal distances on the scale represent equal values in the characteristic being measured.
- It permits comparison of the differences between objects.
- The location of the zero point is not fixed. Both the zero point and the units of measurement are arbitrary.
- Any positive linear transformation of the form $y = a + bx$ will preserve the properties of the scale.
- It is not meaningful to take ratios of scale values.
- Statistical techniques that may be used include all of those that can be applied to nominal and ordinal data, and in addition the arithmetic mean, standard deviation, and other statistics commonly used in marketing research.

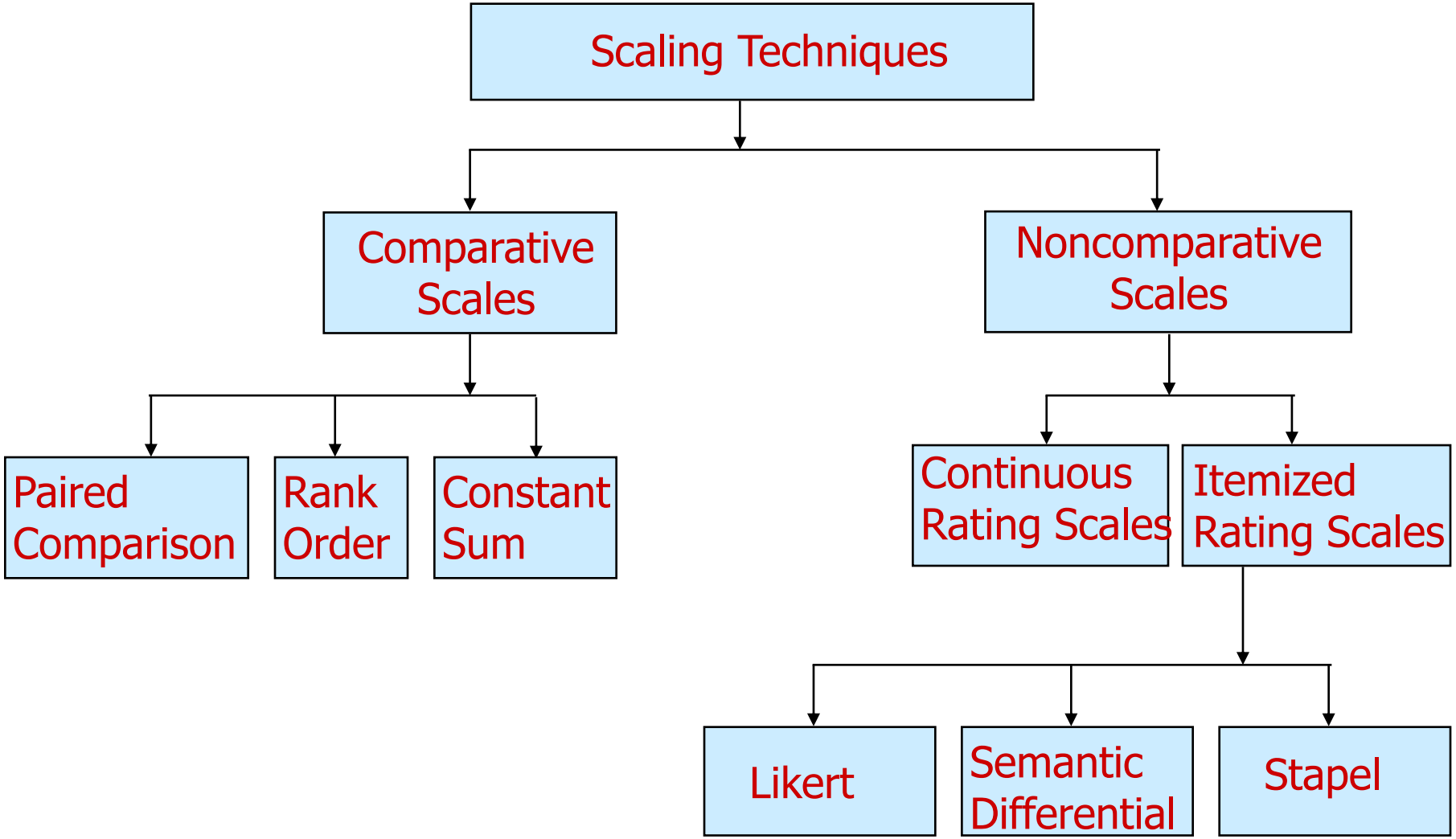
Ratio Scale

- Possesses all the properties of the nominal, ordinal, and interval scales.
- It has an absolute zero point.
- It is meaningful to compute ratios of scale values.
- Only proportionate transformations of the form y = bx, where b is a positive constant, are allowed.
- All statistical techniques can be applied to ratio data.

Primary Scales of Measurement

Scale	Basic Characteristics	Common Examples	Marketing Examples	Permissible Statistics	
				Descriptive	Inferential
Nominal	Numbers identify & classify objects	Social Security nos., numbering of football players	Brand nos., store types	Percentages, mode	Chi-square, binomial test
Ordinal	Nos. indicate the relative positions of objects but not the magnitude of differences between them	Quality rankings, rankings of teams in a tournament	Preference rankings, market position, social class	Percentile, median	Rank-order correlation, Friedman ANOVA
Interval	Differences between objects	Temperature (Fahrenheit)	Attitudes, opinions, index	Range, mean, standard	Product-moment
Ratio	Zero point is fixed, ratios of scale values can be compared	Length, weight	Age, sales, income, costs	Geometric mean, harmonic mean	Coefficient of variation

A Classification of Scaling Techniques



A Comparison of Scaling Techniques

Comparative scales involve the direct comparison of stimulus objects. Comparative scale data must be interpreted in relative terms and have only ordinal or rank order properties.

In **non-comparative scales**, each object is scaled independently of the others in the stimulus set. The resulting data are generally assumed to be interval or ratio scaled.

Relative Advantages of Comparative Scales

- Small differences between stimulus objects can be detected.
- Same known reference points for all respondents.
- Easily understood and can be applied.
- Involve fewer theoretical assumptions.
- Tend to reduce halo or carryover effects from one judgment to another.

Relative Disadvantages of Comparative Scales

- Ordinal nature of the data
- Inability to generalize beyond the stimulus objects scaled.

Comparative Scaling Techniques Paired Comparison Scaling

- A respondent is presented with two objects and asked to select one according to some criterion.
- The data obtained are ordinal in nature.
- Paired comparison scaling is the most widely-used comparative scaling technique.
- With n brands, [n(n - 1) /2] paired comparisons are required.
- Under the assumption of transitivity, it is possible to convert paired comparison data to a rank order.

Rank Order Scaling

- Respondents are presented with several objects simultaneously and asked to order or rank them according to some criterion.
- It is possible that the respondent may dislike the brand ranked 1 in an absolute sense.
- Furthermore, rank order scaling also results in ordinal data.
- Only (n - 1) scaling decisions need be made in rank order scaling.

Constant Sum Scaling

- Respondents allocate a constant sum of units, such as 100 points to attributes of a product to reflect their importance.
- If an attribute is unimportant, the respondent assigns it zero points.
- If an attribute is twice as important as some other attribute, it receives twice as many points.
- The sum of all the points is 100. Hence, the name of the scale.

Non-comparative Scaling Techniques

- Respondents evaluate only one object at a time, and for this reason non-comparative scales are often referred to as monadic scales.

- Non-comparative techniques consist of continuous and itemized rating scales.

Continuous Rating Scale

Respondents rate the objects by placing a mark at the appropriate position on a line that runs from one extreme of the criterion variable to the other. The form of the continuous scale may vary considerably.

How would you rate D-mart as a retail store?

Version 1

Probably the worst - - - - - - -I - Probably the best

Version 2

Probably the worst - - - - - - -I --Probably the best

0 10 20 30 40 50 60 70 80 90 100

Version 3

Very bad Neither good nor bad Very good

Probably the worst - - - - - - -I ---Probably the best

0 10 20 30 40 50 60 70 80 90 100

Itemized Rating Scales

- The respondents are provided with a scale that has a number or brief description associated with each category.
- The categories are ordered in terms of scale position, and the respondents are required to select the specified category that best describes the object being rated.
- The commonly used itemized rating scales are the Likert, semantic differential, and Stapel scales.

Likert Scale

The **Likert scale** requires the respondents to indicate a degree of agreement or disagreement with each of a series of statements about the stimulus objects.

	Strongly disagree	Disagree	Neither agree nor disagree		Agree Strongly
1. D-mart sells high quality merchandise.	1	2X	3	4	5
2. D-mart has poor in-store service.	1	2X	3	4	5
3. I like to shop at D-mart.	1	2	3X	4	5

agree

The analysis can be conducted on an item-by-item basis (profile analysis), or a total (summated) score can be calculated.

When arriving at a total score, the categories assigned to the negative statements by the respondents should be scored by reversing the scale.

Semantic Differential Scale

The **semantic differential** is a seven-point rating scale with end points associated with bipolar labels that have semantic meaning.

D-Mart is:

Powerful --:--:--:--:-X-:--:--: Weak

Unreliable --:--:--:--:--:-X-:--: Reliable

Modern --:--:--:--:--:--:-X-: Old-fashioned

The negative adjective or phrase sometimes appears at the left side of the scale and sometimes at the right.

This controls the tendency of some respondents, particularly those with very positive or very negative attitudes, to mark the right- or left-hand sides without reading the labels.

Individual items on a semantic differential scale may be scored on either a -3 to +3 or a 1 to 7 scale.

Stapel Scale

The **Stapel scale** is a unipolar rating scale with ten categories numbered from -5 to +5, without a neutral point (zero). This scale is usually presented vertically.

D-Mart

+5	+5
+4	+4
+3	+3
+2	+2X
+1	+1
HIGH QUALITY	POOR SERVICE
-1	-1
-2	-2
-3	-3
-4X	-4
-5	-5

The data obtained by using a Stapel scale can be analyzed in the same way as semantic differential data.

Basic Noncomparative Scales

Scale	Basic Characteristics	Examples	Advantages	Disadvantages
Continuous Rating Scale	Place a mark on a continuous line	Reaction to TV commercials	Easy to construct	Scoring can be cumbersome unless computerized
Itemized Rating Scales				
Likert Scale	Degrees of agreement on a 1 (strongly disagree) to 5 (strongly agree) scale	Measurement of attitudes	Easy to construct, administer, and understand	More time-consuming
Semantic Differential	Seven - point scale with bipolar labels	Brand, product, and company images	Versatile	Controversy as to whether the data are interval
Stapel Scale	Unipolar ten - point scale, - 5 to +5, without a neutral point (zero)	Measurement of attitudes and images	Easy to construct, administer over telephone	Confusing and difficult to apply

Data Analysis-Interpretation and Report Writing

WHY DO WE ANALYZE DATA?

- The purpose of analyzing data is to obtain usable and useful information. The analysis, irrespective of whether the data is qualitative or quantitative, may:
- describe and summaries the data
- identify relationships between variables
- compare variables
- identify the difference between variables
- forecast outcomes

DATA ANALYSIS

Data analysis involves processing and analyzing the data to derive meaningful insights, while data interpretation involves making sense of the insights and drawing conclusions.



Data Analysis Techniques

Data analysis techniques involve processing and analyzing the data to derive meaningful insights.
The choice of data analysis technique depends on the research question and objectives.
Some common data analysis techniques are:

1. **Descriptive Statistics-**Descriptive statistics involves summarizing and describing the data using measures such as mean, median, and standard deviation.
2. **Inferential Statistics-** involves making inferences about the population based on the sample data. This technique involves hypothesis testing, confidence intervals, and regression analysis.
3. **Content Analysis** Content analysis involves analyzing the text, images, or videos to identify patterns and themes
4. **Data Mining** Data mining involves using statistical and machine learning techniques to analyze large datasets and identify patterns.



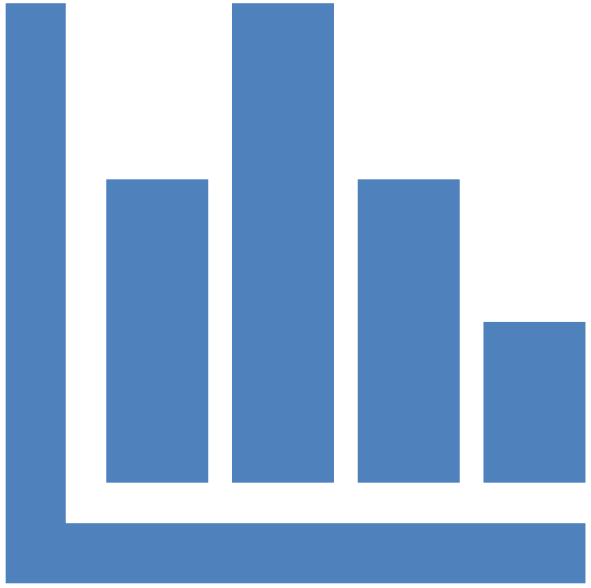
Data Interpretation Techniques

Data interpretation involves making sense of the insights derived from the data analysis.

The choice of data interpretation technique depends on the research question and objectives.

Some common data interpretation techniques are:

1. **Data Visualization** involves presenting the data in a visual format, such as charts, graphs, or tables, to communicate the insights effectively
2. **Storytelling** Storytelling involves presenting the data in a narrative format, such as a story, to make the insights more relatable and memorable.
3. **Comparative Analysis** involves comparing the research findings with the existing literature or benchmarks to draw conclusions.



Data Presentation Techniques

Data presentation involves presenting the data concisely to communicate the research findings. The choice of data presentation technique depends on the research question and objectives. Some common data presentation techniques are:

a. **Tables and Graphs-** Tables and graphs are effective data presentation techniques for presenting numerical data.

b. **Infographics-** Infographics are effective data presentation techniques for presenting complex data in a visual and easy-to-understand format.

c. **Data Storytelling-**Data storytelling involves presenting the data in a narrative format to communicate the research findings effectively.



Hypothesis Testing

Hypothesis testing is a statistical method used to make decisions about a population based on a sample of data. Hypothesis testing is a crucial aspect of conducting high-quality research, as it allows researchers to test their assumptions and draw conclusions based on the data.

Importance of Hypothesis Testing in Research

Hypothesis testing is important in research for several reasons:

Testing Assumptions

Hypothesis testing allows researchers to test their assumptions and draw conclusions based on the data. By testing their assumptions, researchers can avoid making inaccurate or misleading conclusions.

Making Informed Decisions

Hypothesis testing allows researchers to make informed decisions based on the data. By making informed decisions, researchers can ensure that their research findings are accurate, reliable, and valid.

Replicability

Hypothesis testing allows researchers to replicate their research findings. By replicating their research findings, researchers can ensure that their research is valid and reliable.

Basic Concepts concerning Hypothesis Testing

1. **Null Hypothesis:** The null hypothesis is a statement that assumes that there is no significant difference between the sample data and the population data.
2. **Alternative Hypothesis:** The alternative hypothesis is a statement that assumes that there is a significant difference between the sample data and the population data.
3. **Significance Level:** The significance level is a measure of the strength of evidence against the null hypothesis. The significance level is usually set at 0.05 or 0.01.

Procedure for Hypothesis Testing

1. **Define the Null and Alternative Hypotheses The** first step in hypothesis testing is to define the null and alternative hypotheses. The null hypothesis assumes that there is no significant difference between the sample data and the population data, while the alternative hypothesis assumes that there is a significant difference.

2. **Choose a Significance Level** The significance level is a measure of the strength of evidence against the null hypothesis. The significance level is usually set at 0.05 or 0.01. The choice of significance level depends on the research question and objectives.

3. **Calculate the Test Statistic** The test statistic is a measure of the difference between the sample data and the population data. The choice of test statistic depends on the research question and objectives. Common test statistics include t-tests, ANOVA, and chi-square tests.

4. **Calculate the P-Value** The p-value is the probability of obtaining the observed sample data assuming that the null hypothesis is true. The p-value is compared to the significance level to determine whether to reject or fail to reject the null hypothesis.

5. **Interpret the Results** The final step in hypothesis testing is to interpret the results. If the p-value is less than the significance level, then the null hypothesis is rejected in favor of the alternative hypothesis. If the

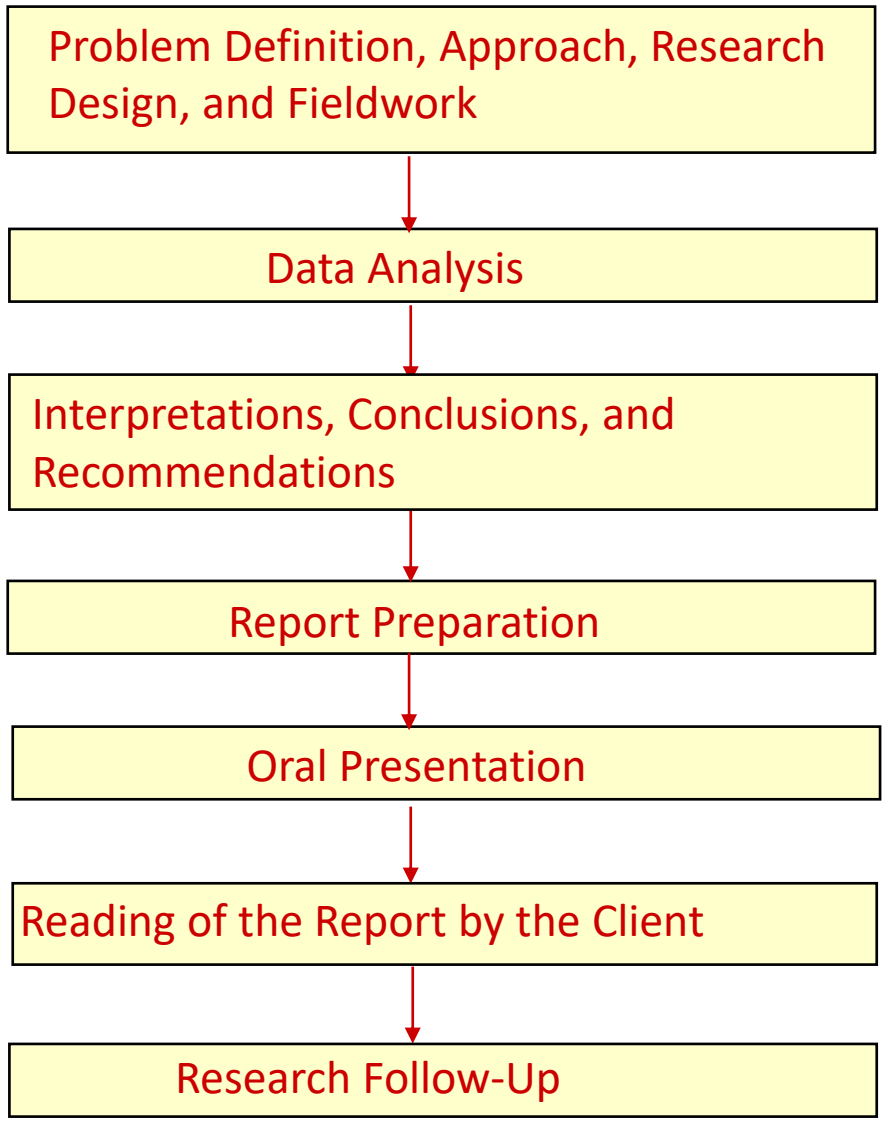
Research Report Writing

Importance of the Report and Presentation

- For the following reasons, the report and its presentation are important parts of the marketing research project:
- After the project is complete and management have made the decision, there is little documentary evidence of the project other than the written report. The report serves as a historical record of the project.
- Management decisions are guided by the report and the presentation.
- The involvement of many decision makers in a research project is limited to reading the written report and listening to the oral presentation. These managers evaluate the quality of the entire project on the quality of the report and presentation.
- Decision makers' decision to undertake marketing research in the future or to use the particular research supplier again will be influenced by the perceived usefulness of the report and the presentation.

The Report Preparation and Presentation Process

Report formats are likely to vary with the researcher or the marketing research firm conducting the project.



Report Format

I. Title page
II. Letter of transmittal
III. Letter of authorization
IV. Table of contents
V. List of tables
VI. List of graphs
VII. List of appendices
VIII. List of exhibits
IX. Executive summary
    a. Major findings
    b. Conclusions
    c. Recommendations

Report Format

X. Problem definition
   a. Background to the problem
   b. Statement of the problem
XI. Approach to the problem
XII. Research design
   a. Type of research design
   b. Information needs
   c. Data collection from secondary sources
   d. Data collection from primary sources
   e. Scaling techniques
   f. Questionnaire development and pretesting
   g. Sampling techniques
   h. Fieldwork

Report Format

XIII. Data analysis
    a. Methodology
    b. Plan of data analysis
XIV. Results
XV. Limitations and caveats
XVI. Conclusions and recommendations
XVII. Exhibits
    a. Questionnaires and forms
    b. Statistical output
    c. Lists

A Good Report Writing

Readers. A report should be written for a specific reader or readers: the marketing managers who will use the results.

Technical jargon should be avoided unless the researcher is sure of the technical abilities and demands of the readers of a report. As expressed by one expert, 'The readers of your reports are busy people; and very few of them can balance a research report, a cup of coffee, and a dictionary at one time.'12 Instead of technical terms such as maximum likelihood, heteroscedasticity and non-parametric, researchers should try to use descriptive explanations. If some technical terms cannot be avoided, definitions should be presented in a glossary or appendix.

Easy to follow. The report should be easy to follow. It should be structured logically and written clearly. The material, particularly the body of the report, should be structured in a logical manner so that the reader can easily see the inherent connections and linkages. Headings should be used for different topics, and subheadings for subtopics. A logical organization also leads to a coherent report.

Presentable and professional appearance. *Presentable and professional appearance.* The look of a report is important. The report should be professionally reproduced with quality paper, typing and binding for hard copies, and with skilful graphic design for online reports. Variation in type size and skilful use of white space can greatly contribute to the appearance and readability of the report. However, a balance should be sought with styles of variation. Too much variation can lead to confusion; variation is only useful if it aids understanding.

Objective Researchers can become so fascinated with their project that they overlook their 'objective' role. The report should accurately present the research design, results and conclusions of the project, without slanting the findings to conform to the expectations of management. Decision makers are unlikely to receive with enthusiasm a report that reflects unfavourably on their judgement or actions. Yet the researcher must have the courage to present and defend the results objectively

Reinforce text with tables and graphs. It is important to reinforce key information in the text with tables, graphs, pictures, maps and other visual devices. Visual aids can greatly facilitate communication and add to the clarity and impact of the report. Guidelines for tabular and graphical presentation are discussed later.

Conciseness. A report should be concise. Anything unnecessary should be omitted. If too much information is included, important points may be lost. Avoid lengthy discussions of common procedures. However, brevity should not be achieved at the expense of completeness.

Guidelines for Tables

Title and number. Every table should have a number (1a) and title (1b).

Arrangement of data items. The arrangement of data items in a table should emphasize the most significant aspect of the data.

Basis of measurement. The basis or unit of measurement should be clearly stated .

Leaders, rulings, spaces. *Leaders*, dots or hyphens used to lead the eye horizontally, impart uniformity and improve readability (4a). Instead of ruling the table horizontally or vertically, white spaces (4b) are used to set off data items. Skipping lines after different sections of the data can also assist the eye. Horizontal rules (4c) are often used after the headings.

Explanations and comments: **Headings, stubs, and footnotes**. Designations placed over the vertical columns are called headings (5a). Designations placed in the left-hand column are called stubs (5b). Information that cannot be incorporated in the table should be explained by footnotes (5c).

Sources of the data. If the data contained in the table are secondary, the source of data should be cited (6a).

RESEARCH & ETHICS

Dr.Rajkumar Rayabaram

Research Ethics

- **Ethics**
- Derived from a Greek word Ethos which means;
  - Custom
  - Convention
  - Spirit of community
- Refers to the moral principles that govern a person's behaviour
- It's a branch of philosophy that deals with morality
- It's concerned with distinguishing between;
  - Good and evil
  - right and wrong human action
  - acceptable and unacceptable behaviour
  - virtuous and non-virtuous characteristics of people

Research Ethics

- **Research Ethics**
  - Set of principles governing the way a research should be designed, managed and conducted
  - Provides guidelines for the responsible conduct of research.
  - Educates and monitors scientists conducting research to ensure a high ethical standard

Why is Research Ethics Important?

- Protect the vulnerable group and other study participant
- Participants are safeguarded from exploitation
- It ensures sufficient knowledge is shared by all concerned
- Establishes risk-benefit ratio for the study subjects
- It ensure respect, privacy, disclosure and fair treatment of subjects/participants
- Build capability of subjects to accept or reject participation
- Promote the aims of research, such as knowledge, truth, and avoidance of error
- Promote the values for collaborative work
- Build public support for research

General ethical principles applied to research with human subjects.

- **Principle of Autonomy**
  - Refers to the right of an individual to determine what activities to or not participate in
  - Individual should be able to understand what is being asked and make a reasoned judgment about the effect of participation
  - The choice to participate should be devoid of coercive influence
- **Principle of Beneficence**
  - Refers to the obligation of the investigator to maximize benefits for the participants/society while minimizing risk of harm
- **Principle of Justice**
  - Requires that there is equitable selection of participants
  - It emphasizes that the subjects involved in research are beneficiaries of the findings of the research

Ethical Issues in Research

- **Voluntary participation**
  - Subjects should be allowed to exercise free will in deciding whether to participate in research
  - participants have rights to withdraw from the study at any stage if they wish to do so
- **Informed consent**
  - Requires open and honest communication between the researcher and the study participant
  - The researcher should providing sufficient information to the subjects on the implications of participation.
  - Subjects' acceptance to participate in the study should be free of coercion
  - Either written or verbal

Ethical Issues……Cont'd

- **Research language**
  - Offensive, discriminatory, or other unacceptable language needs to be avoided in the formulation of items for the research instrument
- **Honesty**
  - Regarded as the truthfulness or accuracy of the researchers actions
  - It involves;
    - Ensuring integrity in the entire process of undertaking the research
    - Commitment to intellectual honesty and personal responsibility for ones actions
  - Example; fabrication, falsification, plagiarism etc

Ethical Issues……Cont'd

- **Confidentiality**
  - Involves protecting information obtained from the participants
  - Identity of subjects should be concealed eg use of study codes
- **Respect for the participants**
  - Concerned with the wellbeing of the subjects
  - Examples
    - Assuring privacy and confidentiality
    - Sharing results
    - Providing emerging information etc

Ethical Issues……Cont'd

- **Social responsibility**
  - Seeks to promote social good
  - Prevent or mitigate social harms through research, public education, and advocacy
- **Objectivity**
  - This is the state or quality of being true and without individual biases, interpretations, feelings, and imaginations
  - The researcher should seek to reduce or eliminate biases, prejudices, or subjective evaluations by relying on verifiable data

THANK YOU

Research is a never-ending marathon

- **Dr.Rajkumar Rayabaram**